Concept Drift in media bias classification

Machine Learning Methods for Data Streams

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Outline

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Introduction

Modern online media can be treated as a **stream of textual data**, where political articles are published continuously and may reflect changing narratives, framing strategies, and ideological language patterns over time.

The aim of this project is to develop and evaluate two complementary approaches for **drift detection in political news streams**:

1. **Part I**: Classification-based drift detection using online classifiers
2. **Part II**: Unsupervised distributional drift detection

Dataset: GDELT articles (Oct 2024 – Apr 2026), 890 articles for binary left/right classification, 4200 articles in total.

Synthetic Data Generation

To validate drift detection under **controlled conditions**, we created a synthetic data generator with ideologically-coded vocabularies:

Component	**Left**	**Right**
Class vocabulary (V_L, V_R)	50 terms	50 terms
Shared neutral vocabulary	130 terms	
Cross-contamination	10%	
Neutral noise	40%	

Drift simulation types:

- **Abrupt**: Instantaneous vocabulary swap
- **Gradual**: Distributed swap over time window
- **Recurring**: Vocabulary restoration

Part I: Classification-Based Drift

Approach: Online classifiers process articles sequentially with test-then-train evaluation.

Two vectorization strategies:

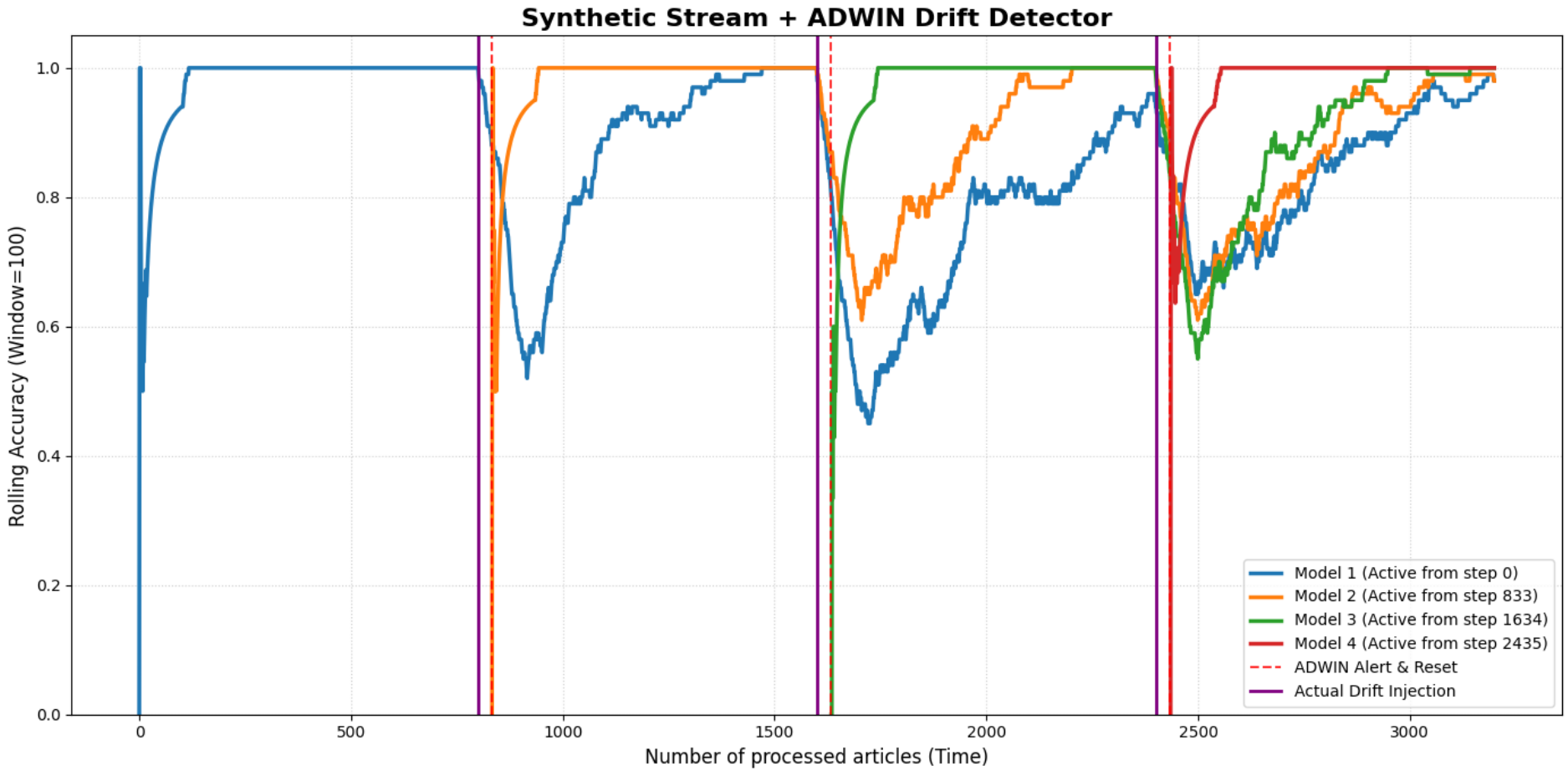
1. **TF-IDF (sparse)**: Lightweight bag-of-words baseline from `river`
2. **LLM Embeddings (dense)**: all-MiniLM-L6-v2 Sentence Transformer (384-dim)

Classifiers:

- TF-IDF pipeline: MultinomialNB
- LLM pipeline: LogisticRegression (SGD)
- Baseline: Majority-class prediction (rolling 50-sample window)

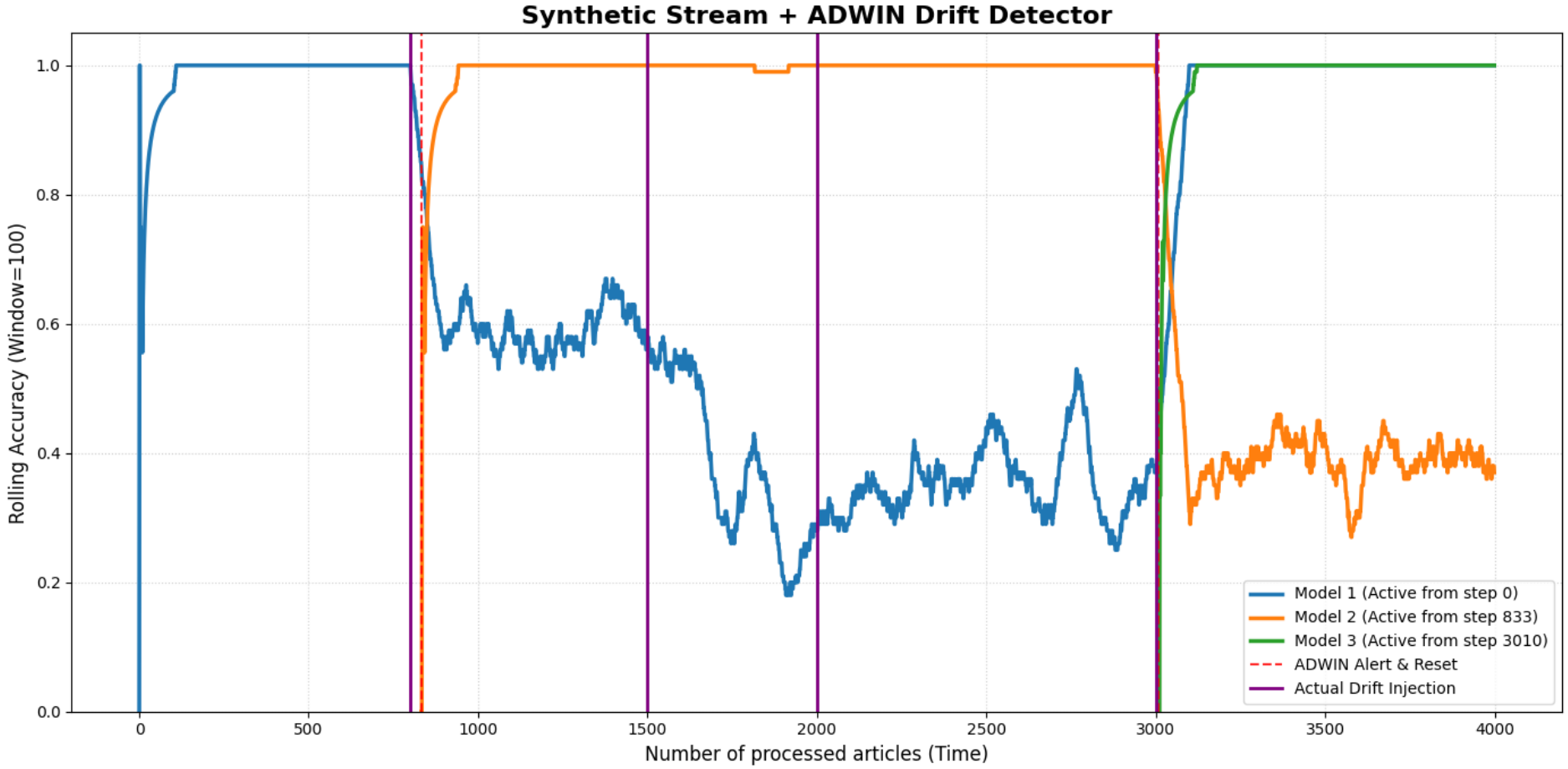
Drift detection: ADWIN

Synthetic Results: Basic Drift



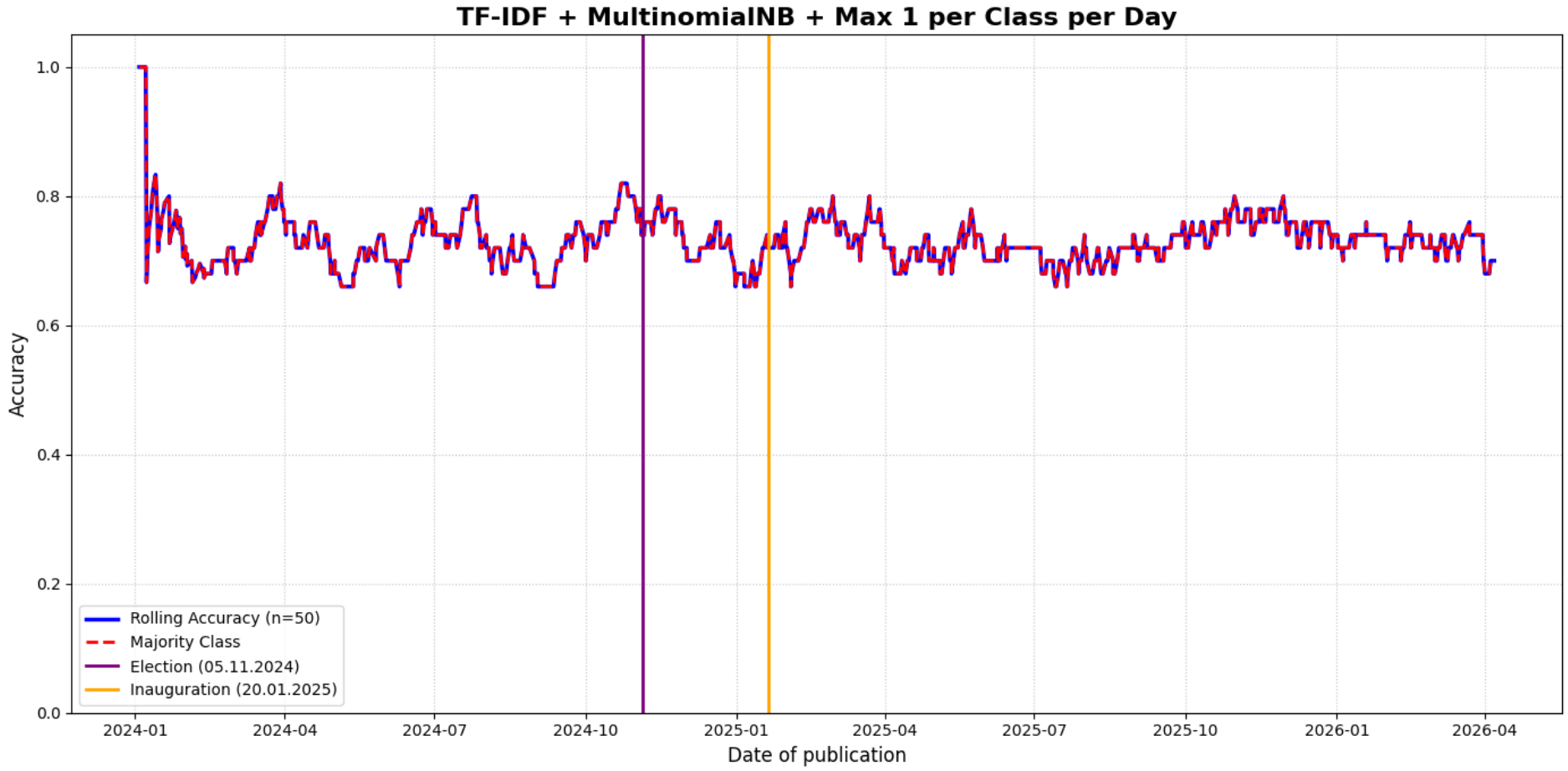
Key finding: ADWIN drift detector successfully identifies all three injected drifts. Fresh models recover to near-100% accuracy within hundreds of steps.

Synthetic Results: Advanced Drift



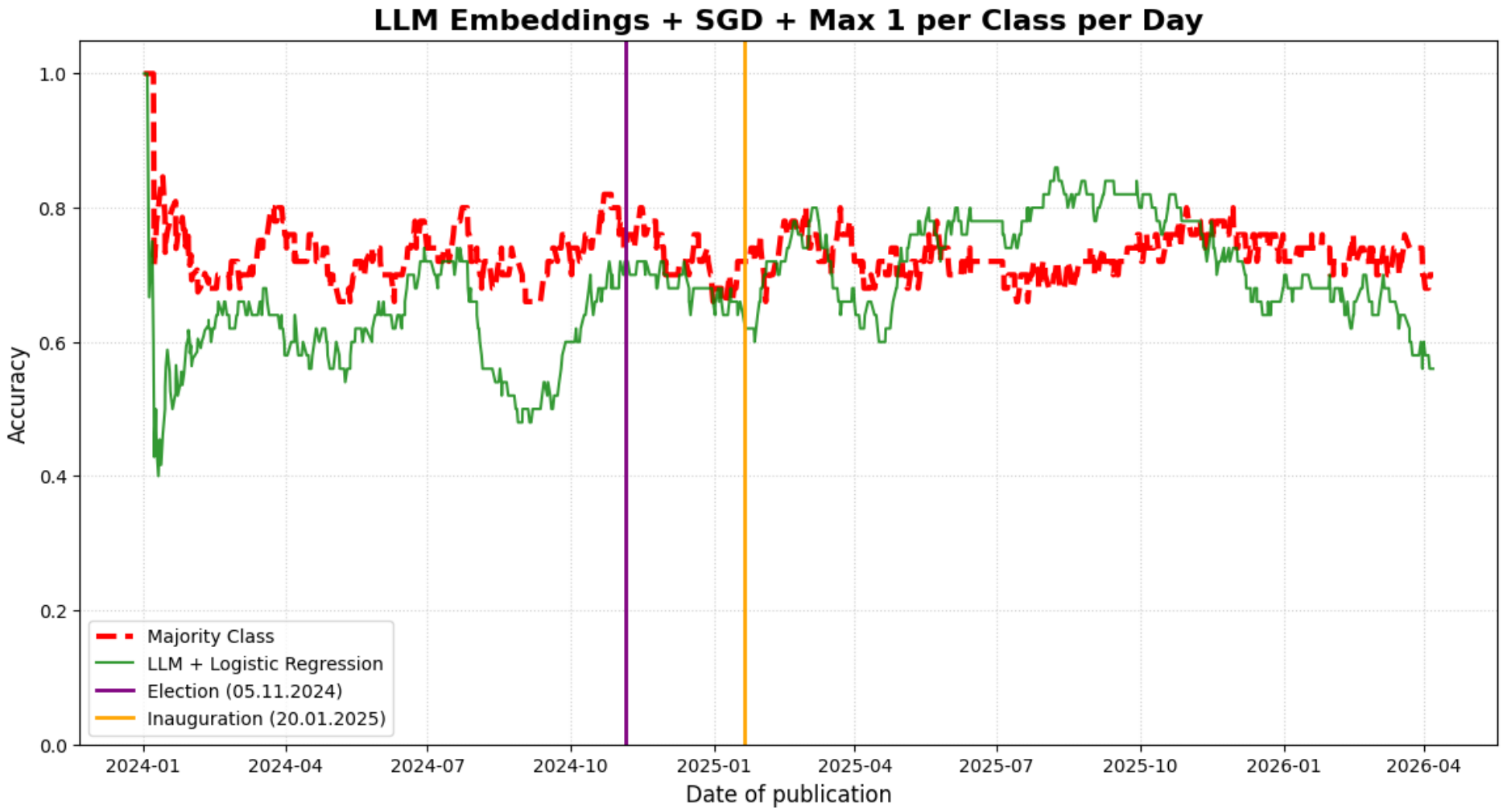
Key finding: Abrupt and recurring drifts trigger ADWIN alerts. Gradual drift was not detected.

Real-World Results: TF-IDF



Critical finding: Model accuracy overlaps perfectly with majority-class baseline.

Real-World Results: LLM Embeddings



Finding: Dense embeddings also perform at baseline level with no improvement over majority-class prediction.

Part II: Distributional Drift Detection

Approach: Abandon requirement for accurate classification. Instead, detect drift by measuring semantic shifts directly.

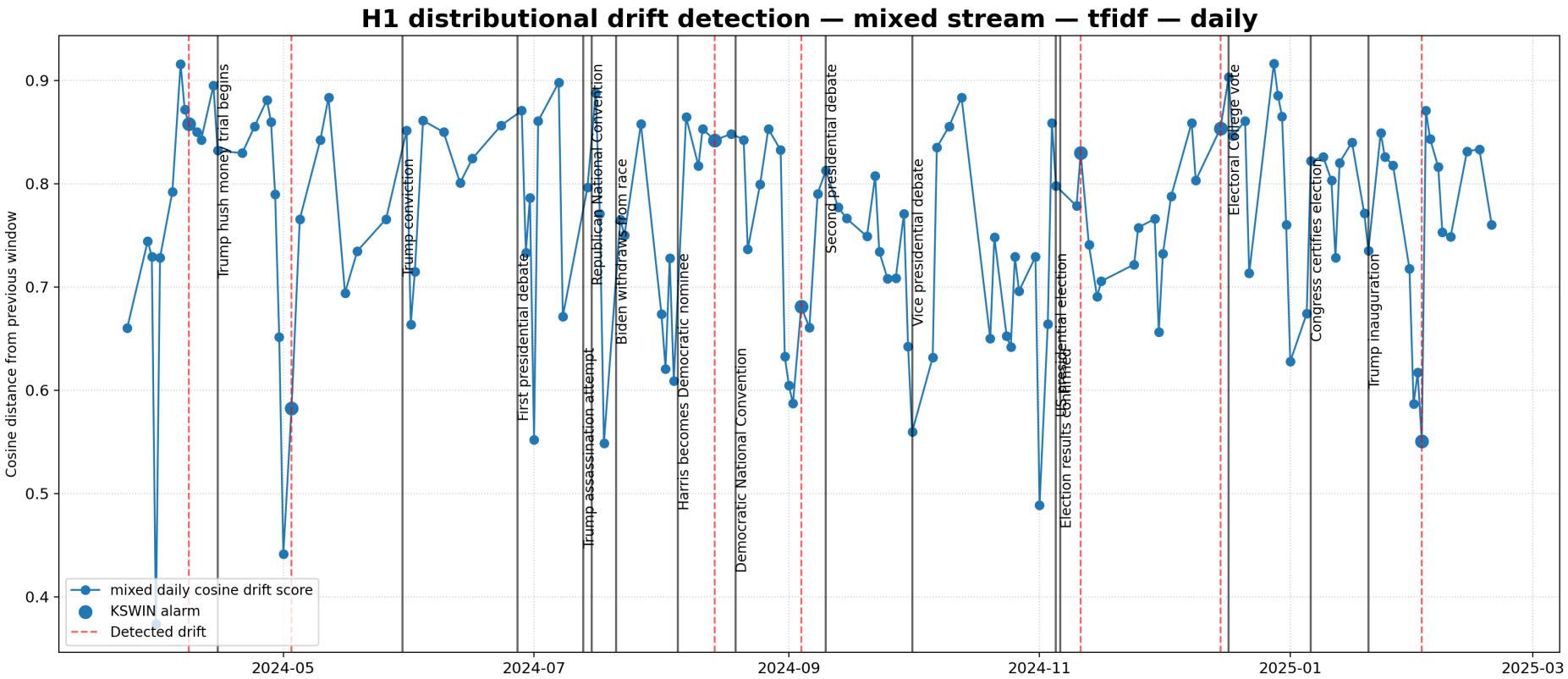
Pipeline:

1. **Aggregation**: Group articles into daily/weekly windows
2. **Vectorization**: Embed aggregated documents
3. **Distance stream**: Compute cosine distance between consecutive windows
4. **Drift detection**: Apply ADWIN, PageHinkley, and KSWIN to distance stream
5. **Event matching**: Validate detections against known political events ($\pm$7–14 days)

Advantages:

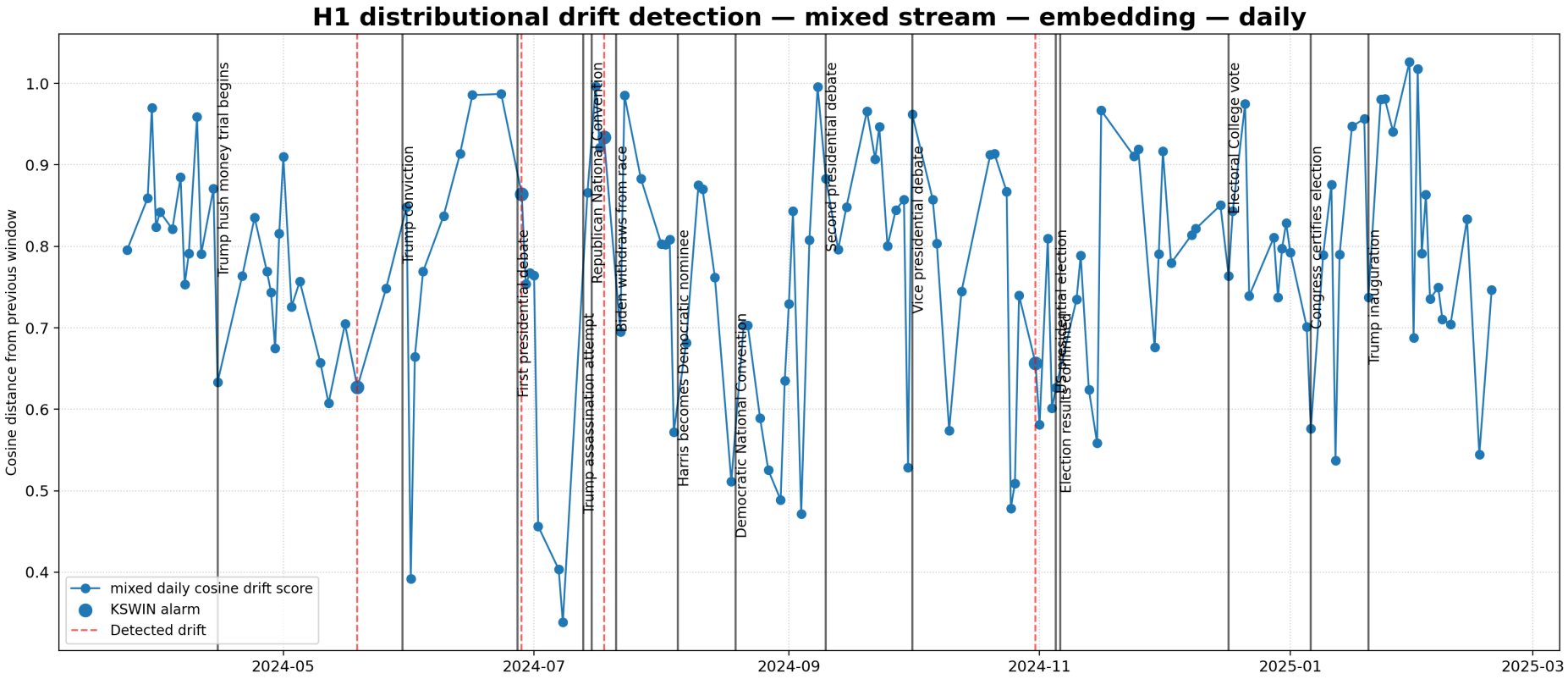
- No dependency on classification accuracy
- Directly measures semantic shifts in media discourse
- Multiple detectors improve robustness

Part II Results: TF-IDF (Mixed Stream)



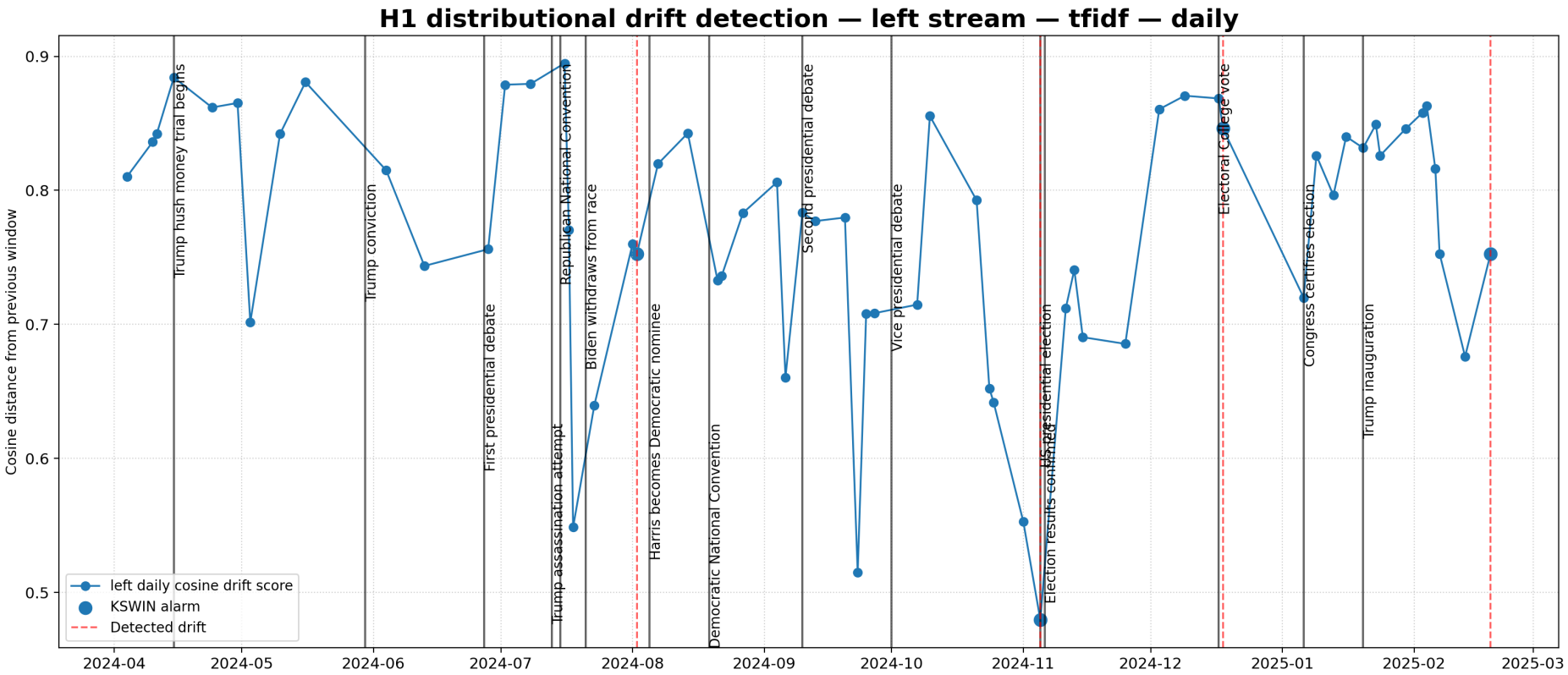
Observation: TF-IDF produces clean drift detection with moderate event alignment around conventions and election period.

Part II Results: Embeddings (Mixed Stream)



Observation: Dense embeddings exhibit noisier signals.

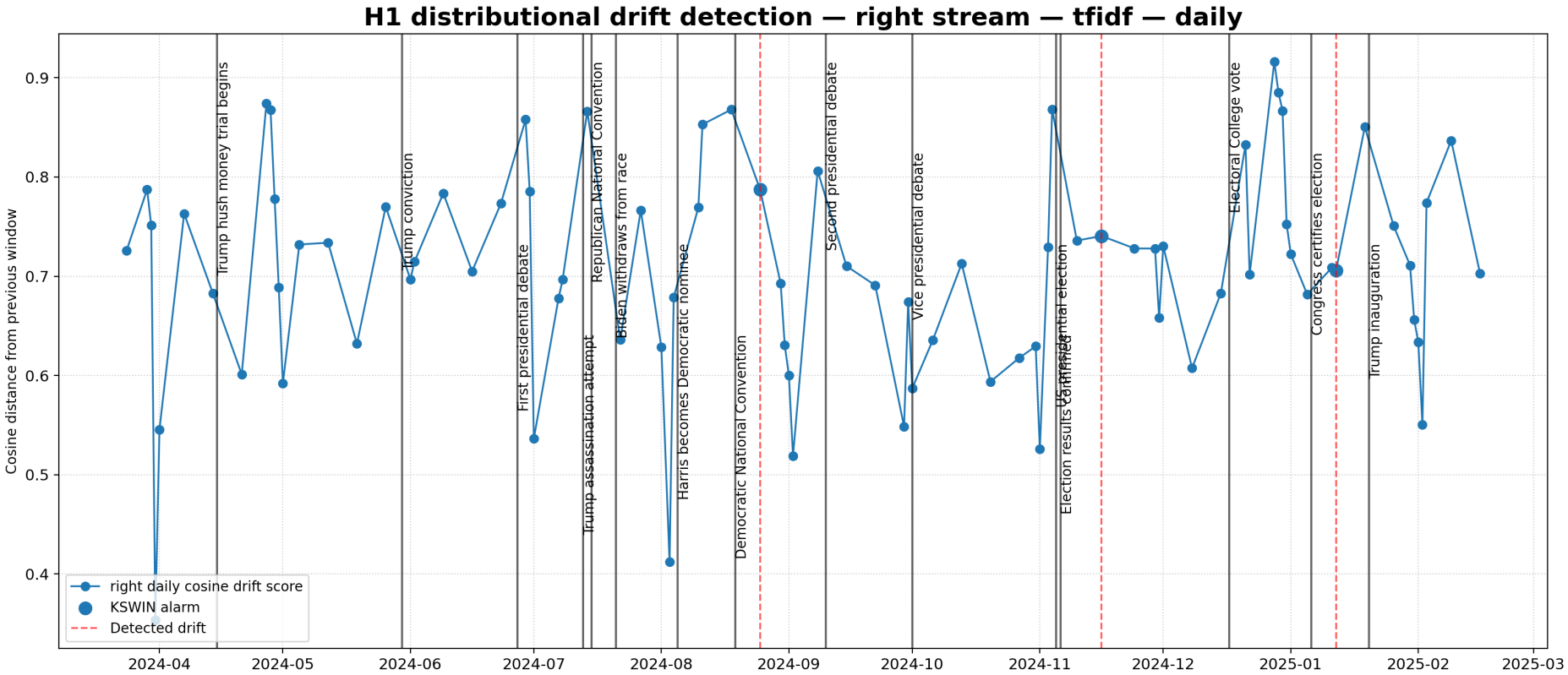
Part II Results: Left-Leaning Stream



Key finding: Sharp drift signals aligned with Democratic-specific events:

- Biden's withdrawal (July 21, 2024)
- Harris nomination (August 5, 2024)

Part II Results: Right-Leaning Stream



Key finding: Pronounced drifts synchronized with Republican-specific events:

- Trump conviction (May 30, 2024)
- Trump assassination attempt (July 13, 2024)

Summary of Findings

Part I: Classification-Based Drift Detection

Fails on real-world data. Both TF-IDF and LLM approaches collapse to majority-class prediction, providing zero discriminative signal despite reliable performance on synthetic data.

Part II: Distributional Drift Detection

Succeeds by abandoning classification. TF-IDF outperforms embeddings. Ideology-specific stream analysis reveals clear event-drift alignment.

Limitations & Future Work

Limitations:

- Small corpus for classification
- Binary left/right categorization is oversimplified
- LLM embeddings not fine-tuned on political discourse

Future directions:

- Larger, more diverse article corpus with multiple outlets per ideology
- Domain-specific fine-tuned embeddings (RoBERTa on political text)

Thank you for your attention!

Any questions?